Here are good notes for your exam, arranged by Block, Unit, and topics, drawing directly from the provided sources and highlighting important points for clarity.

Block 1: Research Process and Concepts

Unit 1: Basic Process/Concept in Research

- **Introduction to Research**
    - Research is a **scientific quest** for understanding the human mind, behavior, and mental processes.
    - Psychologists, as scientists, systematically answer questions about thinking, feeling, and behaving, develop principles, and use them to solve problems.
    - It is a process through which **new knowledge is discovered**.
    - Conducting research requires following a sequence of steps, which may vary slightly based on the type of research (quantitative or qualitative).
- **Definition and Meaning of Research**
    - The word "research" combines "re" (again, a new) and "search" (to examine closely and carefully, to test and try, or to probe).
    - It describes a **careful, systematic, patient study and investigation** in some field of knowledge, undertaken to establish facts or principles.
    - The simplest meaning is to search for facts, answers to research questions, and solutions to problems.
    - **Scientific Research** is a **systematic and objective attempt** to provide answers to certain questions, aiming to discover and develop an organized body of knowledge.
    - Definitions from literature:
        - Encyclopaedia of Social Science: "the manipulation of generalising to extend, connect or verify knowledge…".
        - Kerlinger (1973): "systematic, controlled, empirical and critical investigation of hypothetical propositions about the presumed relationship about various phenomena".
        - Burns (1994): "a systematic investigation to find answers to a problem".
    - Therefore, research refers to a **systematic method** involving problem enunciation, hypothesis formulation, data collection and analysis, and reaching conclusions (solutions or theoretical generalizations).
- **Criteria of Good Research**
    - **Purpose** should be **clearly defined**, and common concepts **operationally defined**.

    - Procedure should be **precisely planned, focused, and appropriately described** for replication and advancement.
    - Research design should be **carefully planned to maintain objectivity**.
    - Research report should be **frank** to gauge findings' effects.
    - Data analysis must be **adequate to reveal significance**, and the method of analysis **appropriate**.
    - **Validity and reliability** of data should be **carefully examined**.
- **Objectives of Good Research**

    - To discover answers to questions through scientific procedures, finding hidden truths.
    - Broad groupings of objectives:
        - To gain familiarity with a phenomenon or achieve new insights (exploratory research).
        - **To portray accurately** characteristics of an individual, situation, or group (descriptive research).
        - To determine **frequency** of occurrence or association (diagnostic research).
        - **To test a hypothesis** of a causal relationship between variables (hypothesis-testing/experimental studies).
- **Qualities of a Good Research**

    - **Systematic**: Structured by rules, following steps in sequence, involves creative thinking, avoids guessing/intuition.
    - **Empirical**: Conclusions based on **hardcore evidence** from real-life experiences and observations, providing external ability to results.
    - **Valid and Verifiable**: Involves precise observation and accurate description, uses reliable/valid instruments, results are correct and can be verified by others.
    - **Logical**: Guided by rules of reasoning, logical processes of induction (specific to general) and deduction (general to specific), makes research feasible and meaningful.
    - **Develops Theories and Principles**: Helps in accurate prediction, allows sound generalizations about populations beyond immediate observations.
    - **Replicable**: Designs, procedures, and results should be replicable by others to assess validity and allow reuse/transmission of results.
- **Research Process: Basic Considerations**

    - Psychology is scientific because it uses scientific method principles and practices.
    - Psychological research process has two major categories:
        - **Context of Discovery**: **Initial phase** where observations, beliefs, information, and general knowledge lead to **new ideas or ways of thinking** about phenomena.
            - **Role of Theories, Hypotheses, and Paradigms**:

        - Researchers assume **determinism** (all events result from specific causal factors) and that behavior/mental processes follow discoverable patterns.
        - **Psychological theories** attempt to understand brain, mind, behavior, and environment, using interrelated principles to explain/predict phenomena.
        - **Hypothesis**: A **tentative and testable explanation** of the relationship between two or more events or variables. A **variable** is any factor that changes in size or quality (e.g., mood, test performance).
        - **Paradigm**: A **model of functions and interrelationships** of a process, a "way of thinking" about the world and how to study it. Paradigms can shift and lead to knowledge revolutions.
    - **Research Biases**: Challenges to objectivity, distort impressions of data.
        - **External influences**: Culture, media influencing worldview.
        - **Personal bias**: Distorts evaluation due to personal beliefs, attributes, or past experiences.
        - **Observer bias**: Taking some events as meaningful and others not.
        - **Expectancy bias**: Researchers expecting specific outcomes may see what they expect, rather than remaining objective, making observed events seem "discovered" instead of "created" by expectations.
        - **Placebo biases**: Occur when people strongly want to believe a treatment is successful, leading to perceived positive outcomes (e.g., sugar pill).
- **Context of Justification**: The **second phase** where results are prepared for communication with other scientists.
    - Relies on the **scientific method** (general procedures for gathering/interpreting evidence to limit errors and yield dependable conclusions).
    - **Scientific Attitudes and Values**: Curiosity, critical and skeptical attitude towards conclusions until repeatedly duplicated by independent investigations. Secrecy is banned, data and methods must be open for **public verifiability**.
    - **Objectivity Safeguards**:
        - **Procedural safeguards**: Keeping complete records in understandable form for evaluation, publishing scientific reports in similar forms.
        - **Standardisation**: **Using uniform, consistent procedures** in all phases of data collection (e.g., same instructions, treatment for all subjects).

        - **Operationalisation**: **Standardising the meaning of concepts** by defining them in terms of how they are measured or produced.
        - **Avoiding bias**: Using various control procedures to avoid external, personal, observer, and expectancy biases.
- **Steps in Research Process** (series of actions for smooth scientific research)

    - **Step I: Identification of the Problem**:
        - First and most important step: asking a question or identifying a need from curiosity.
        - Psychological studies focus on questions like: what causes behavior? what is the nature of behavior? how are internal processes linked with behavior?.
        - Involves understanding the problem thoroughly and rephrasing it meaningfully, deciding what to find out.
        - Evaluate problem against funds, time, expertise, and knowledge available.
        - Identify factors to examine (simple like age, complex like violent cartoons' effects).
        - Factors should: not have been investigated, contribute to understanding, be available to investigate, and potentially lead to new questions.
        - **Review of Literature**: Undertaken to identify a good solvable problem.
            - Helps eliminate duplication, provides guidance, discovers/selects relevant variables, estimates previous work, systemizes knowledge, and redefines variables.
            - Sources include journals, books, abstracts, indexes, periodicals (e.g., PsycINFO, Psychological Abstracts).
    - **Step II: Formulating a Hypothesis**:
        - After problem identification and literature review, a hypothesis (suggested answer) is formulated.
        - Plays a **key role in guiding any study**.
        - Derived from earlier research, existing theories, personal observations/experiences.
        - **Tentative statement showing a relationship between variables under study**, stated as a declarative sentence (e.g., "Those who are rewarded shall require lesser number of trials to learn the lesson than those who are not rewarded").
        - Must be formulated **in advance of data-gathering** for unbiased research.
    - **Step III: Identifying, Manipulating and Controlling Variables**:
        - **Variables**: Characteristics manipulated, controlled, and observed by the experimenter.
        - **Dependent Variable (DV)**: The characteristic or condition that changes as the experimenter changes the independent variables; the variable about which prediction is made (e.g., speed of learning).

        - **Independent Variable (IV)**: The condition or characteristic manipulated or selected by the experimenter to find its relationship to observed phenomena (e.g., meaningfulness of material, noise level).
            - Type E IV: manipulated directly by the experimenter (e.g., noise level).
            - Type S IV: selected by the experimenter (e.g., age of workers).
        - **Extraneous Variable**: Uncontrolled variable that may affect the dependent variable, not of primary interest but must be controlled (e.g., age of smoker, diet habits in smoking study).
        - **Operational Definitions** are crucial for complex psychological variables, defining concepts in terms of measurable operations (e.g., verbal, behavioral, physiological measures).
    - **Step IV: Formulating a Research Design**:
        - A **blueprint of procedures** for testing the relationship between IV and DV.
        - Selection depends on research purpose, variable types, and experimental conditions.
        - Main purpose: **manipulate IV freely and provide maximum control of extraneous variables** to ensure experimental change is due to IV manipulation.
        - Explains how research questions will be answered, sets out logic of inquiry.
        - Includes logistical arrangements, measurement procedures, sampling strategy, analysis frame, and time frame.
        - Crucial for valid findings/conclusions; faulty design leads to misleading findings.
        - **Constructing Device for Observation and Measurement**: Choosing/constructing tools like **questionnaires and interview schedules**.
        - **Sample Selection and Data Collection**: Deciding on study participants (sample representing population), choosing specific data collection methods (observation, experiment, case study, survey), and administration logistics.
    - **Step V: Data Analysis and their Interpretation**:
        - Collected data analyzed using quantitative/statistical and qualitative techniques.
        - Purpose: make sense of data, understand relation to problem and hypotheses, draw conclusions.
        - Can use univariate, bivariate, or multivariate analysis.
        - Statistical analysis often aims to **reject the null hypothesis** to accept the alternative hypothesis.
    - **Step VI: Drawing Conclusion**:
        - Investigator draws statements about the research problem based on analyzed results.

        - **Hypotheses are accepted or rejected**.
        - Conclusions related to original theory or findings; theory may need modification based on new findings.
    - **Step VII: Preparation of Report and Publication**:
        - Last step, documenting all research steps clearly.
        - Informs what was done, discovered, and concluded.
        - Helps readers understand, use, and **replicate the study**.
        - Publication in scientific journals/books for wider dissemination.
- **Importance of Research in Psychology**

    - Well-recognized in human life, leading to progress in organizational behavior, medical sciences, education.
    - Helps gauge human and animal behavior through empirical and theoretical studies (e.g., learning, motivation).
    - Practical gains: improved treatment methods, better vehicle designs, enhanced worker performance/happiness.
    - Psychological research is highly scientific, often using experimental methods, leading to control and manipulation of variables.
    - Applies rigorous scientific norms and statistical methods for data collection, organization, description, and analysis, making it method-oriented.

Unit 2: Reliability and Validity (External and Internal)

- **Introduction**

    - Most research aims to draw cause-and-effect conclusions and develop theories.
    - Reliability and validity are **central issues in all measurement**, concerning the connection of measures to constructs.
    - Both help establish the **truthfulness, credibility, or believability of findings**.
- **Reliability**

    - **Meaning**: Any significant results **must be repeatable** by other researchers under same conditions.
    - It is the **consistency of your measurement**, or the degree to which an instrument measures the same way each time it is used under the same conditions with the same subjects.
    - In short, it is the **repeatability of measurement**.
    - A measure is reliable if a person's score on the same test given twice is similar (e.g., a neuroticism test should yield similar results each time).
    - Reliability is **estimated**, not measured.
    - Error of measurement implies some inaccuracy in psychological measurement, and the goal is to minimize this error.
- **Methods of Estimating Reliability**

- **External Consistency Procedures**: Compare findings from two independent data collection processes.
    - **Test-Retest Reliability**:
        - Most frequently used method.
        - Repeating the **same test on the same sample** at two different times.
        - Reliability coefficient is the **correlation between scores** from both administrations.
        - Assumes no substantial changes in the construct measured.
        - **Time gap** is critical; shorter gap usually means higher correlation.
        - Limitations:
            - **Memory Effect/Carry Over Effect**: Subjects remember responses if administrations are too close, leading to artificial reliability (e.g., pre-test/post-test).
            - **Practice Effect**: Improved scores due to repeated testing (e.g., IQ tests).
            - **Absence**: People missing re-tests.
    - **Parallel Forms Reliability** (Alternate forms, Equivalent form, Comparable form reliability).
        - Compares **two equivalent forms of a test** that measure the same attribute, using different items but same item selection rules.
        - Administered to the same group, sometimes on the same day or at different times.
        - Pearson product moment correlation coefficient used as reliability estimate.
        - Sources of variation: random error and differences between forms (if same day); also time sampling error (if different times).
        - Provides one of the most **rigorous assessments of reliability**, but less common in practice due to difficulty of creating two forms.
- **Internal Consistency Procedures**: Items measuring the same phenomena should produce similar results.
    - **Split-Half Reliability**:
        - Randomly divides all items measuring the same construct into two sets.
        - Complete instrument administered, total scores calculated for each half.
        - Reliability is the **correlation between these two scores**.
        - Problem: shorter tests may lose reliability; more useful for long tests.
        - **Spearman-Brown's formula** can be used to rectify shortness and estimate correlation as if each part were full length: $r = (2rhh)/(1 + rhh)$.
    - **Kuder-Richardson Estimate of Reliability (KR-20)**:
        - Coefficient of internal consistency.

        - Relevant when each test item is scored **0 or 1** (e.g., right or wrong answers).
        - Formula: KR-20 = [N/(N-1)] * [1 - (Σpq/σ2)], where N=items, σ2=total test variance, p=proportion correct, q=proportion wrong.
    - **Cronbach's Alpha (α)**:
        - May be thought of as the **mean of all possible split-half coefficients**, corrected by the Spearman-Brown formula.
        - Extends Kuder-Richardson method to tests with items not scored as 0 or 1 (e.g., attitude scales with Likert responses).
        - Varies between 0.00 and 1.00.
- **Comparison of Reliability Estimators**
    - Each has pros and cons.
    - **Inter-rater** (not explicitly defined in sources as a main category but implied by "multiple observers"): Best for measures involving observation, requires multiple observers or repeated ratings by a single observer.
    - **Parallel forms**: Best for two alternate measures of the same thing.
    - **Internal consistency measures** (split-half, alpha, KR-20): Require multiple items designed to measure the same construct.
    - **Cronbach's Alpha**: Useful when many items are present.
    - **Test-retest reliability**: Mostly used in experimental/quasi-experimental designs, depends on control group availability.
    - Test-retest and inter-rater estimates are generally lower than parallel forms and internal consistency due to different measurement times or raters.
- **Validity**
    - **Meaning**: The degree to which a test **measures what it claims to measure**.
    - Essential for proper administration and interpretation of a test.
    - Inferences drawn from the test must be **appropriate, meaningful, and useful**.
    - Requires comparison with some **ideal independent measures or criteria**.
    - **Validity coefficients**: Correlation coefficients between the test and an ideal criterion.
- **Types of Validity**
    - **Content Validity**:
        - The notion that a test should **sample the range of behavior** represented by the theoretical concept being tested.
        - Ensures items represent the entire range of possible items the test should cover (e.g., spelling test for third graders).
        - Inbuilt through careful item selection; expert judges can rate item relevance for hard-to-define traits.
    - **Criterion-Related Validity**:
        - A valid test should **relate closely to other measures** of the same theoretical concept.

        - Demonstrated when a test effectively predicts a criterion or indicators of the construct.
        - Two types:
            - **Concurrent Validity**: Criterion measures obtained **at the same time as test scores**. Reflects the degree to which test scores estimate an individual's **present status** regarding the criterion (e.g., anxiety test reflecting current anxiety level).
            - **Predictive Validity**: Criterion measures obtained **at a time after the test**. Useful for identifying future success/failure (e.g., aptitude tests for entrance exams).
    - **Construct Validity**:
        - Complex type; property of a test that the measurement **actually measures the constructs they are designed to measure**.
        - Ways to determine:
            - Test should measure its theoretical construct, not something else (e.g., leadership test should not test extraversion).
            - Test should not measure theoretically **unrelated constructs** (e.g., music aptitude test not requiring too much reading ability).
            - Test should **predict results related to theoretical concepts** it measures (e.g., music ability test predicting who benefits from lessons).
        - Two types:
            - **Convergent Validity**: Extent to which a measure is **correlated with other measures theoretically predicted to correlate with it**.
            - **Discriminant Validity** (Divergent validity): Extent to which the operationalisation is **not correlated with other operationalisations that it theoretically should not be correlated with**.
    - **Face Validity**:
        - Refers to **what appears to measure superficially**.
        - Depends on the **judgment of the researcher**; items are scrutinized until satisfied they accurately measure the construct.
        - Based on subjective opinion.
    - **Internal Validity**:
        - The **most fundamental type of validity**; concerns the **logic of the relationships between the independent and dependent variables**.
        - An estimate of the degree to which inferences about **causal relationship can be drawn**.
        - Achieved through properly suited experimental techniques with controlled observation of IV effect on DV.
        - **Threats to Internal Validity**: Factors that can undermine the conclusion that the independent variable caused changes in the dependent variable.

        - **Confounding**: Error when effects of two variables cannot be separated, leading to confused interpretation (e.g., age confounded with IV).
        - **Selection bias**: Bias in selecting groups, pre-test differences interact with IV (e.g., gender, personality differences between experimental and control groups).
        - **History**: Events outside the experiment or between measures influencing participants' responses (e.g., natural disasters during study).
        - **Maturation**: Subjects changing during the experiment (e.g., growth, fatigue, illness) altering their reaction to IV.
        - **Repeated testing**: Participants biased by remembering answers or conditioning from incessant testing.
        - **Instrument change**: If measurement instruments are replaced/changed, affecting results.
        - **Regression toward the mean**: Tendency of extreme scores on a first measure to be closer to the mean on a second testing (e.g., low ability students improving not due to intervention, but statistical artifact).
        - **Mortality**: Participants dropping out of the study, leading to bias between groups.
        - **Diffusion**: Treatment effects spreading from treatment groups to control groups, reducing observed differences.
        - **Compensatory rivalry/resentful demoralization**: Control groups changing behavior due to knowledge of the study (e.g., working harder or less hard).
        - **Experimenter bias**: Experimenters unintentionally behaving differently to experimental vs. control groups, affecting results. Reduced by blinding experimenters and standardizing procedures.
    - **External Validity**:
        - Concerns whether **results can be generalized to another situation** (different subjects, settings, times, etc.).
        - Lacks when small samples from particular locations or with idiosyncratic features are used, limiting applicability to broader populations.
        - **Threats to External Validity**: Limitations on generalization, especially when cause (IV) depends on other factors.
            - **Aptitude-Treatment-Interaction**: Sample features (e.g., volunteers, specific demographics) limit generalization.
            - **Situations**: Situational factors (treatment conditions, light, noise, location, experimenter, timing, measurement scope/degree) may limit generalizations.
            - **Pre-Test Effects**: Cause-effect relationships only found after pre-tests, limiting generality.

        - **Post-Test Effects**: Cause-effect relationships only explored after post-tests, limiting generalizations.
        - **Rosenthal Effects**: Derivations from cause-consequence relationships cannot be generalized to other investigators/researchers.

Unit 3: Variables and Constructs

- **Meaning of Variables**
    - A **variable** is simply **something that varies**.
    - It is a **property that takes different values**.
    - Values of a variable must be **observable** (e.g., outcome of dice roll has six observable values).
    - Attributes of objects, events, things, and beings that can be **measured** (e.g., number of items solved, speed of response, IQ, sex, anxiety level, illumination).
    - **Variables are tangible** (duration, frequency, items checked), while **theoretical concepts are intangible** (hunger, motivation, anxiety).
    - Variables are related to theoretical concepts via **operational definitions** used to measure the concepts.
- **Types of Variables**
    - **Stimulus, Organism, and Response Variables (S-O-R Model)**:
        - Psychologists use this model to study behavior or its causes.
        - **S (Stimulus variables)**: Some form of energy in the environment to which an organism is sensitive (e.g., electrical shock intensity).
        - **O (Organism variables)**: Changeable physiological and psychological characteristics of the observed organisms (e.g., anxiety level, age, heart rate, skin resistance).
        - **R (Response variables)**: Some behavior or action of the organism (e.g., pressing a lever, reaction time, latency to jump).
    - **Independent and Dependent Variables**:
        - **Independent Variable (IV)**: The factor **manipulated or selected** by the experimenter to ascertain its relationship to an observed phenomenon.
            - **Type E independent variable**: Directly manipulated by experimenter (e.g., continuous noise, intermittent noise, no noise).
            - **Type S independent variable**: Selected based on existing characteristics (e.g., age of workers).
        - **Dependent Variable (DV)**: The factor that **appears, disappears, or varies as the experimenter introduces, removes, or varies the independent variable**. It is a measure of the subject's behavior.
            - Dimensions of measurement: **frequency**, **duration**, **latency**, **force**.
        - The relationship is one of **dependence**; DV depends on IV (e.g., speed of learning depends on meaningfulness).

- **Extraneous and Confounded Variables**:
  - **Extraneous Variables**: Any other variables that may **'mask' the relationship between IV and DV**. They may directly affect the DV or combine with the IV to produce an effect.
    - Must be **controlled** to determine if DV changes due to IV.
    - Types: **organismic variables** (subject-related: age, sex, intelligence), **situational variables** (environmental: noise, temperature), and **sequential variables** (sequence effects: adaptation, fatigue from repeated conditions).
  - **Confounding**: Occurs when two variables get combined so that the **effect of one cannot be separated from the effect of the other**. Makes study conclusions doubtful.
    - Example: Studying TV viewing on violence perception, if experimental group is adolescents and control group is adults, age is confounded with TV viewing.
- **Active and Attribute Variables**:
  - **Active Variables**: Any variable that is **manipulated** by the experimenter (e.g., reward, punishment, teaching methods).
  - **Attribute Variables**: Variables that are **not manipulated but measured** by the experimenter; human characteristics (e.g., intelligence, aptitudes, sex, socio-economic status).
- **Quantitative and Categorical Variables**:
  - **Quantitative Variables**: Varies in **amount** (e.g., speed of response, intensity of sound, IQ). Precise and accurate measurement possible as they can be ordered by magnitude.
  - **Categorical Variables**: Varies in **kind** (e.g., gender, race, religion).
    - **Constant**: Only one value (e.g., taxi, tree).
    - **Dichotomous**: Only two categories (e.g., yes/no, good/bad).
    - **Polytomous**: More than two categories (e.g., religion, political parties, attitudes on a scale).
- **Continuous Variables and Discrete Variables**:
  - **Continuous Variable**: Capable of being measured in **any arbitrary degree of fineness or exactness** (e.g., age, height, intelligence, reaction time).
  - **Discrete Variables**: Not capable of being measured in arbitrary fineness because they contain a **clear gap**; countable (e.g., number of family members, number of books in a library).

- **Constructs**

  - A **construct** is a **concept** that has an added meaning: it is **invented or adopted for a special scientific purpose**.
  - They are **part of a theoretical framework** and related to other constructs.
  - Usually **operationally defined** to allow observation and measurement (e.g., "intelligence" defined by a specific test).

    - Example: "Reinforcement" is a construct related to drive, motivation, and can be operationally defined as "any stimulus or event which increases the probability of the occurrence of a (desired) response".
- **Types of Constructs**
    - **Intervening Variables**:
        - A construct used as a **summary term for a group of other constructs**.
        - Has **no meaning apart from the context** in which it is used.
        - Internal, **directly unobservable psychological processes** that are inferred from behavior (e.g., "reaction potential" as a combination of "habit strength" and "drive"; "hostility" inferred from hostile acts).
    - **Hypothetical Constructs**:
        - A **theoretical term employed to describe something "real"**.
        - An intermediary that has **tangible characteristics**.
        - Defined directly by the operations that established them or by which they were measured (e.g., "habit strength" defined as number of reinforced trials; "reflex" referring to chain of events within an organism).

Unit 4: Hypothesis Formulation and Sampling

- **Introduction**
    - Hypotheses are **tentative explanations and potential answers to a problem**.
    - They provide direction and help researchers interpret data.
- **Meaning and Characteristics of Hypothesis**
    - An important consideration after formulating a research problem.
    - A **tentative and testable statement of the possible relationship between two or more events/variables** under investigation.
    - Needs to be stated in a way that allows **empirical testing**.
    - **Characteristics of a good hypothesis**:
        - **Conceptually clear**.
        - Must be **testable**.
        - Should be related to **existing body of theory and impact**.
        - Should have **logical unity and comprehensiveness**.
- **Formulation of Hypothesis**
    - Science proceeds with observation, hypothesis formulation, and hypothesis testing (accept or reject).
    - A specific hypothesis **narrows the focus of data collection** and helps design the procedure.
    - It's essential for directing and interpreting results.
    - Usually **derived from the problem statement**.
    - Should be formulated in a **positive and substantive form before data are collected**.

    - A creative task involving thinking, imagination, and innovation.
    - **Context of discovery** (how a scientist arrives at hypothesis) vs. **context of justification** (logical reconstruction of ideas for communication).
    - Good hypotheses are "made, not born" by applying methodological criteria.
    - Points to consider: **expected relationship/differences**, **operational definition of variables**, formulated **following literature review**.
    - Can be put in **"if...then" statements** (e.g., "if A is true then B should follow").
- **Possible Difficulties in Formulation of a Good Hypothesis**
    - **Absence of knowledge of a theoretical framework**.
    - Lack of awareness of **available detailed theoretical evidences**.
    - Lack of awareness of **scientific research techniques**.
- **Types of Hypotheses**
    - **Null Hypothesis (Ho)**:
        - Useful tool in testing the significance of difference.
        - Asserts that there is **no true difference** between two population means, or **zero relationship** between terms of the hypothesis (e.g., "schizophrenics and normal do not differ with respect to digit span memory").
        - An important component of inferential statistics.
        - If differences are significant, the null hypothesis is rejected.
        - Should always be a **specific hypothesis**.
        - Example: Ho: μHV ≤ μLV (mean of high vocabulary group is less than or equal to mean of low vocabulary group).
    - **Alternative Hypothesis (H1 or Ha)**:
        - Specifies those values that the **researcher believes to hold true**, and hopes sample data will lead to its acceptance.
        - Represents all other possibilities and indicates the **nature of the relationship**.
        - Example: H1: μHV > μLV (mean of high vocabulary group is greater than mean of low vocabulary group).
        - Can be **directional** (predicts specific direction, e.g., higher/lower) or **non-directional** (predicts only that groups will differ, e.g., μ1 ≠ μ2).
- **Errors in Testing a Hypothesis**
    - Can occur due to faulty sampling, inaccurate data collection, faulty study design, inappropriate statistical methods, or incorrect conclusions.
    - **Type I Error**: **Rejection of a null hypothesis when it is true**.
    - **Type II Error**: **Acceptance of a null hypothesis when it is false**.
- **Importance of Hypothesis Formulation**
    - **Basic function of scientific research**.
    - Provides **focus and direction** to the investigation.

    - Without it, research can be unfocused and results lack clear meaning.
    - **Links between theory and investigation**, leading to new knowledge.
- **Sampling**
    - **Definition**: The process of **selecting units from a population of interest** so that studying the sample can fairly generalize results back to the population.
    - Researchers typically collect data from a sample rather than the entire population due to high cost and dynamic nature of populations.
    - A **miniature picture or smaller representative** of a larger whole.
- **Sampling Terminology**
    - **Population (Universe)**: A **well-defined set of all elements** pertaining to a given characteristic; includes all observations/measurements of a characteristic. Can be finite (countable) or infinite (unlimited).
    - **Parameter**: A measure based upon the **entire population**.
    - **Sample**: Any number of persons selected to represent the population according to a plan; a **smaller representation of the population**.
    - **Statistic**: A measure based upon a **sample**.
    - **Sample size (n)**: Number of selected individuals/units.
    - **Sampling Design/Strategy**: The techniques or procedures adopted for selecting sampling units from which inferences about the population are drawn.
    - **Sampling Unit/Element**: Each individual or case that becomes the basis for selecting a sample.
    - **Sampling Frame**: The **list of people from which the sample is taken**; should be comprehensive, complete, and up-to-date (e.g., Electoral Register).
- **Purpose of Sampling**
    - To derive desired information about the population at **minimum cost or with maximum reliability**.
    - To achieve **maximum precision** in estimates within a given sample size and to **avoid bias** in selection.
    - Bias can occur if: non-random method is used, sampling population not covered accurately, or a section is impossible to find/refuses to cooperate.
- **Sampling Methods** (Two main categories)
    - **Non-Probability Sampling Methods**:
        - **No way of assessing the probability** of an element being included in the sample.
        - Provide no basis for estimating how closely sample characteristics approximate population parameters.
        - Do not use random sampling.
        - **Types**:
            - **Haphazard/Accidental/Convenience Sampling**: Researcher selects **convenient cases**; can produce ineffective, highly

unrepresentative samples; cheap and quick but prone to systematic errors (e.g., person-on-the-street interviews).
        - **Quota Sampling**: Researcher identifies **relevant categories** (e.g., male/female, age groups) and decides how many to get in each, then uses haphazard/convenience sampling within categories. Ensures some differences, but still prone to bias as interviewers may choose easily accessible people. Less costly and more convenient than some other methods.
        - **Purposive Sampling**: Uses the **judgment of an expert** in selecting cases, or selects cases with a **specific purpose** (e.g., exploratory or field research). Researcher never knows if selected cases represent population, but useful for selecting **unique, informative cases**. Less costly, more accessible, convenient, but no guarantee of representativeness.
        - **Snowball Sampling** (Network, Chain Referral, Reputation Sampling): Begins with one or more contacts, who then provide information for other potential respondents. Most useful when **population is unknowable** or there are very few lists (e.g., studying small informal social groups). Limitations: cumbersome for large samples, does not allow probability statistical methods, introduces bias.
        - **Systematic Sampling**: Selecting **every Nth person** from a predetermined list. Possesses some randomness (first element selected randomly) but also non-probability traits (excludes elements between selected ones). Relatively quick and easy to check. Disadvantage: prone to sampling error if list is arranged in a particular order.
    - **Probability Sampling Methods**:
        - **Clearly specify the probability** or likelihood of inclusion of each element.
        - **Free of bias** in selecting sample units.
        - Help in estimating sampling errors and evaluating sample results for precision, accuracy, and efficiency.
        - Conclusions are **generalizable**.
        - **Types**:
            - **Simple Random Sampling**:
                - Requires a **complete listing of all elements**.
                - Each element has an **equal chance of being selected**.
                - Selection of one element has **no effect on the chance of selecting another**.
                - **Sampling with replacement**: Element returned to population after selection.
                - **Sampling without replacement**: Element not returned.

        - All possible combinations of samples of fixed size have an equal probability of being selected.
        - **Advantages**: Each person has equal chance, serves as a foundation, suitable for small populations, more representative with larger sample size, least costly, easily assessable accuracy.
        - **Disadvantages**: Requires complete/up-to-date universe, large sample size for reliability, high cost/time for geographically dispersed populations, risk of wrong results with unskilled investigators.
    - **Stratified Random Sampling**:
        - Population divided into **two or more strata** (subpopulations) based on criteria (e.g., sex, education), which are internally homogeneous and non-overlapping.
        - A **simple random sample** is taken from **each stratum**.
        - **Advantages**: More **representative** of the population, increases **precision** in estimating attributes, provides **convenience** in sampling.
        - **Disadvantages**: Improper stratification causes wrong results, heavy cost/time for geographical concentration, requires trained investigators.
    - **Cluster Sampling** (Multi-stage Sampling):
        - Uses **multiple stages** to cover wide geographic areas.
        - **Aggregated units (clusters) are randomly selected**, and then samples are drawn from these selected clusters.
        - Example: Randomly selecting states, then districts within states, then schools within districts, then children within schools.
        - **More flexible** than other methods, sub-divisions needed only for selected units.
        - **Less accurate** than a simple random sample of the same size.
- **Importance of Sampling**
    - Gained great importance in all types of studies due to vastness of populations, difficulties in contacting people, high refusal rates.
    - **Cost is lowest**, **data collection is faster**.
    - Smaller dataset allows for **homogeneity** and improves **accuracy and quality of data**.

Block 2: Research Methods

Unit 1: Survey Research

- **Introduction**
    - A significant research tool in applied social science, primarily for **collecting data**.
    - Involves asking questions from respondents/subjects.
    - Can range from a short paper-and-pencil feedback form to an in-depth interview.
    - Gathers data about people's thoughts and behaviors using questionnaires or statistical tools.
- **Concept and Meaning of 'Survey Research'**
    - A **non-experimental, descriptive research method**.
    - One of the **quantitative methods** used for studying **large samples**.
    - Researcher collects data using **standardized questionnaires or interviews** administered to a sample from a population.
    - Helpful in collecting data through both **direct** (e.g., face-to-face interview) and **indirect** (e.g., opinions on library services) observation.
- **Steps Involved in Conducting Survey Research** (Systematic process)
    - **Determination of the aims and objectives of study**: Analyze relevant areas, clearly specify focused aims and objectives.
    - **Define the population to be studied**: Define the target population (collection of people/objects with common characteristics).
    - **Design and construct a survey**: Design the survey, select instruments (e.g., telephonic interview), and conduct a **pilot study** (small preliminary survey) to analyze significance/relevance of instruments.
    - **Select a representative sample**: Sample chosen to represent maximum characteristics of the whole population. If not representative, findings cannot be generalized.
    - **Administer the survey**: Conduct the survey using the selected instrument on the sample to collect data.
    - **Analyze and interpret the findings**: Code and process data using statistical tools, then interpret findings.
    - **Prepare the report**: Document all details including aims, objectives, data analysis, interpretation, and discussion; evaluate how findings meet proposed aims.
    - **Communicate the findings**: Disseminate results to target population, record for future research, assess impacts, and recommend policies.
- **Methods/Instruments Used in Collecting Data Through Survey Research** (Three major ways)
    - **Sampling**: The technique of sampling itself can act as an instrument.

        - Can be done with **randomization** (equal chance for each subject, e.g., lottery method) or **stratification** (categorizing population and then sampling).
    - **Questionnaire**: Paper-pencil, multiple-choice tests; data collected from large samples at once.
        - **Mail survey**: Soft copy via mail, inexpensive, less time-consuming, convenient, but difficult for on-the-spot/detailed answers.
        - **Group administered questionnaire**: Large number of respondents in a group setting answer structured questions. Advantage: respondents can clarify doubts instantly.
        - **Household drop-off survey**: Researcher personally delivers and collects questionnaires door-to-door, allowing respondents convenience.
    - **Interview**: Face-to-face interaction allowing follow-up questions, best for opinions/impressions.
        - **Structured interview**: Questions **prepared and preplanned in advance**, imposed serially.
        - **Unstructured interview**: Informal atmosphere, nothing preplanned, questions depend on subject's response.
        - **Telephonic interview**: Saves time and money, but sample limited to those with phones.
- **Types of Survey Research**
    - Can use qualitative (open-ended) or quantitative (forced-choice) measures.
    - **Cross-sectional Surveys**:
        - Collects data from **varied or different types of groups at a single time** (e.g., children of different age groups).
        - Less time-consuming and economical.
    - **Longitudinal Surveys**:
        - Studies the **same sample for a longer period of time** to study behavioral/attitude changes.
        - **Types**:
            - **Trend studies**: Analyze a phenomenon's trend in a population. **Sample might not be the same**, but from the same population, sampled regularly (e.g., yearly survey of graduates using library).
            - **Cohort studies**: Focus on a particular population, sampled and studied more than once within a time gap (e.g., investigating library usage of 2009 graduates in 2009, then again in 2013 with a different sample of 2009 graduates).
            - **Panel studies**: Uses the **same sample of people ("panel") every time** to investigate changes in attitudes, behavior, or practices. More specific and focused. Faces high **attrition rates** (difficulty in availability of same people).
- **Constructing a Survey Research**

    - Needs careful design of framework, content, format, and wordings of survey instruments.
    - Questions should be **clear, specific, relevant, short**, and capable of eliciting response.
    - Considerations: content/scope/objectives of question, response format (e.g., Likert scale), framing questions to elicit response, formatting question series, and being **sensitive to moral values and ethics**.
- **Types of Questions in a Survey Research**

    - **Structured Questions**: **Preplanned and predefined format**.
        - **Dichotomous questions**: Only **two possible responses** (e.g., yes/no, true/false, male/female).
        - **Level of measurement based questions**:
            - **Nominal**: Based on names/classifications (numbers denote serial order, not preference, e.g., categories like General, OBC).
            - **Ordinal**: Based on **ranks and preferences** (e.g., ranking business tycoons).
            - **Interval**: Based on **ratings** (e.g., Likert response scale 1-5 or 1-7).
        - **Filter or Contingency Questions**: Main question followed by succeeding sub-questions, depending on the response to the main question (e.g., "Have you attended the university library? If yes, how many times?"). Should not exceed 2-3 levels to maintain interest.
    - **Unstructured Questions**:
        - Usually used in interviews.
        - Researcher may not prepare a list of questions; series depends on subject's response, or questions asked in an informal atmosphere.
        - Researcher should probe silently, verbally encourage, ask for clarification, and show empathy.
- **Precautions while Designing Instruments**

    - Items/questions should be **clear, specific, relevant, and short**.
    - Respondents must be **capable of answering**.
    - Researcher should have **empathy** and avoid questions with negative impact.
    - Researcher should **avoid any kind of bias**.
- **Advantages and Disadvantages of Survey Research**

    - **Advantages**:
        - Convenient, less time-consuming, economical.
        - Can be conducted for longer periods, allowing insight into changes.
        - Researcher can organize and present reasons for study to get full/honest answers.
    - **Disadvantages**:
        - Maintaining privacy in group interviews can be questionable, restricting honest answers.

        - **High attrition rate** can hinder longitudinal studies.
- **Difficulties and Issues of Survey Research**
    - **Issues on selecting the type of survey**: Choosing appropriate survey type, knowing target population's language, analyzing geographic dispersion.
    - **Issues on constructing the survey**: Suitability, clarity, specificity, and length of questions can be controversial.
    - **Bias Issues**: Researcher's bias and prejudices can significantly influence findings; need to avoid false reports and maintain socially desired behavior.
    - **Administrative Issues**: Cost, mode of survey, feasibility of area, and required time period must be preplanned.

Unit 2: Ex-Post Facto Research

- **Introduction**
    - Used when the researcher wants to study the **cause of an event which has already occurred** as an effect of that cause.
    - Example: Analyzing reasons behind an economic recession that has already happened.
- **Forms of Research Design**
    - **Prospective Research Design**:
        - Researcher tries to find out **future or potential results** (the effect) **after treating the independent variable** (the cause).
        - Examines effects *after* manipulation (e.g., asking a subject to smoke to see if it causes cancer).
    - **Retrospective Research Design**:
        - Researcher tries to **trace the history** in order to find out the basic reason behind an event that **has already occurred**.
        - Examines causes *after* an effect is observed (e.g., tracing reasons for cancer in a person back to smoking/heredity).
        - Ex-post facto research falls under this category.
- **Concept and Meaning of Ex-Post Facto Research**
    - A kind of research where the researcher **predicts the possible causes behind an effect that has already occurred**.
    - Example: Finding reasons for a child's delinquency (lack of discipline, family history, peer effect).
    - Researcher predicts a cause based on a **controlled effect** (effect cannot be varied).
    - Defined as an empirically based investigation where the **independent variable has already occurred** and cannot be manipulated.
    - Conclusions on relationships are inferred without intervening or varying the variables.

    - "Ex-post facto" refers to an experiment where a researcher **examines the effect of a naturally occurring treatment after it has occurred**.
    - A **study that attempts to discover the pre-existing causal conditions between groups**.
    - **Most serious danger**: Concluding causation just because two factors are associated (e.g., delinquency and parenthood).
- **Characteristics of Ex-Post Facto Research** (Also known as 'causal comparative research')

    - Has a **control or comparison group**.
    - The **independent variable (behavior, action, event, treatment) cannot be manipulated or changed** because it has already occurred.
    - Focuses on the **effects** first, then tries to determine the causes.
    - Analyzes the **'how' and 'what' aspects** of an event (how and what reasons led to the phenomena).
    - Explores **possible effects and causes**.
- **Differences between an Experimental and an Ex-Post Facto Research** | Feature | Experimental Research | Ex-post Facto Research | | :---------- | :---------- | :---------- | | **Control over IV** | Researcher can **directly manipulate** the independent variable (cause). | Researcher **cannot directly manipulate** the independent variable (cause). | | **Randomization** | Researcher **can use the principle of randomization** to infer cause-effect. | Researcher **cannot use the principle of randomization** (no direct control over cause). | | **Manipulation** | Researcher **can manipulate variables**. | Researcher **cannot manipulate variables**. | | **Interpretation** | **Easier to interpret** cause-effect relationships. | **Difficult to interpret** relationships due to multiple possible causes for an effect. |

- **Essentials/Requisites for Inferring Causal Relationships** (To conclude cause and effect)

    - **Associative variation**: There should be an **association** between cause 'x' and effect 'y'; a variation in 'x' should yield 'y'.
    - **Systematic order of events**: Cause 'x' must occur **before or simultaneously with** 'y', not after it.
    - **Absence of other causes**: Researcher must analyze all aspects to find the **best possible cause** that led to the effect, excluding other factors (e.g., pinpointing etiology of cancer).
- **Steps of Ex-Post Facto Research** (Systematic sequence)

    - **Determining the problem**: Focus on, analyze, and define the problem to be studied.
    - **Literature Review**: Study related/similar literature and studies for analysis, prediction, and conclusion of causal relationships.

    - **Formulation of hypothesis**: Propose possible solutions/alternatives that might have led to the effect, list assumptions.
    - **Data Collection**: Select sample that fits criteria, select scale/instrument, analyze relationships.
    - **Validity of the research**: Validate significance of findings, ensure findings are valid, significant, and helpful.
    - **Interpretation of the conclusion**: Analyze, evaluate, and interpret collected information to select the best possible alternative cause.
    - Despite its validity, ex-post facto research has shortcomings, and **only partial control is possible**.
- **Strengths and Weaknesses of Ex-Post Facto Research**
    - **Strengths**:
        - Relevant where **variables cannot be manipulated or altered** (e.g., sociological/educational variables like delinquency, achievements).
        - More useful than experimental research for **analyzing broader social variables**.
        - Allows researcher to analyze based on **personal opinion** to reach conclusions.
    - **Weaknesses**:
        - Cannot manipulate independent variables.
        - Cannot randomly assign subjects to different groups.
        - May not provide a **reasonable explanation** for the relationship between IV and DV.
- **Concept of Post Hoc Fallacy**
    - A **bias** where humans tend to arrive at conclusions/predictions that **because two factors go together, one is the cause and the other is the effect**.
    - Example: Concluding delinquency is caused by parenthood, when peer group might be the actual reason.

Unit 3: Experimental Research (Field Experiment)

- **Introduction**
    - Research classified into laboratory experiments, field experiments, field studies, and survey research.
    - This unit focuses on experimental research and field experiments.
- **Experimental Research and Field Experiments**
    - **Identifying the Research Problem**: First step is to define the problem, often requiring **operationalization** (how to measure it), then formulate a hypothesis (positive, negative, or null).
    - **Experimental Research**:
        - Mainly used in **science subjects** (physics, chemistry, medicine).

        - Involves **manipulating an independent variable (IV)** and observing its effect on a **dependent variable (DV)**.
        - All other **extraneous factors are completely controlled within the laboratory**.
        - Systematic and scientific approach using manipulation and controlled testing to understand **causal processes**.
        - **Subjects are homogeneous**.
        - **Controlled conditions**.
        - One experimental group is subjected to IV manipulation.
        - **Cause-effect relationship can be clearly established** (pure effects).
        - **Prediction is possible** and accurate given same conditions.
        - Always **quantitative** in terms of results.
        - **Replicable**.
        - Researcher has **complete control** over variables, random assignment, and scheduling of IVs.
        - Can use **high-level advanced statistical methods** (e.g., F test, correlation, regression).
    - **Field Experiments**:
        - Refer to experiments conducted in **real-life situations**.
        - **Control of extraneous factors is not possible** to the same absolute degree as in a laboratory.
        - Typically use **two groups matched for various factors** (e.g., age, sex, education) in a natural setting.
        - One group subjected to intervention, results compared with the other (control) group.
        - Example: Studying teaching methods in a school by randomly assigning students to experimental (lecture) and control (tutorial) groups and comparing academic performance.
        - Also called **quasi-experiments**.
        - Important for human subjects as it provides **inputs about phenomena in real-life situations**.
        - Subjects may **vary in characteristics**.
        - Subjects are in **natural settings**.
        - Cause-effect relationship can be established **to quite an extent**, but not with same accuracy as lab experiments.
        - Prediction possible **to a certain extent**.
        - Both **qualitative and quantitative** in terms of results.
        - **Replicable but may require modifications** in matching factors.
- **Strengths and Weaknesses of Field Experiments**
    - **Strengths**:
        - Useful for **behavioral and social scientists** (social psychologists, sociologists, educationists).

        - Appropriate for studying **complex social influences, processes, and changes in real-life settings** (e.g., dynamics of small groups).
        - Suited to **testing theory and solving practical problems**.
        - Suited to **testing broad hypotheses**.
        - Offers **flexibility and applicability to a wide variety of problems**.
    - **Weaknesses/Limitations**:
        - Higher chances of **extraneous variables confounding** research findings due to uncontrolled nature.
        - Negative attitude of researcher.
        - Requires **consent and cooperation** from subjects and institutional authorities.
        - Faces **lack of precision problem**.
- **Constructing the Field Experiment** (Various aspects to consider)
    - **Problem identification**: Define problem properly, ensuring it can be researched with appropriate tools.
    - **Sampling**: Proper selection of sample is crucial for systematic research.
        - **Probability samples** (random sampling, e.g., stratified, cluster, systematic, simple random) vs. **Non-probability samples** (no random sampling, e.g., quota, purposive, accidental).
    - **Research design**: A **blueprint of the research work** (setting, subjects, instruments, manipulation, measurement, control group).
        - Must specify number of subjects for experimental and control groups.
    - **Tools of data collection**: What tools will be used, how results measured, what statistical tools.
    - **Procedure**: Obtain permission, select subjects randomly, assign to groups, conduct pre-tests, introduce intervention to experimental group, conduct post-tests, record scores.
    - **Statistical analysis**: Use appropriate statistics (e.g., t-test) to determine significance of differences.
- **Research Design**
    - "Research" means "again + explore".
    - A scientific methodology in a **controlled setting**; observation and experiments are basic scientific tools.
    - **Controlled observation**: Impact of IV on DV observed under specific controlled conditions; IV manipulated systematically, changes in DV recorded.
    - **Kerlinger (1998)** defines research design as the **plan, structure, and strategy of investigation**.
        - **Plan**: Overall scheme or program of the study.
        - **Structure**: More specific, the outline, scheme, or paradigm of variable operation.
        - **Strategy**: Most specific, the method to collect, analyze, interpret data; how objectives will be reached and problems tackled.

    - **Myers (1980)**: Design is the **general structure of the experiments, not its specific content**.
- **Objectives of Research Design** (Two basic objectives)

    - **To provide answers to research questions**: In terms of **validity, objectivity, accuracy, and economical aspects**. Different problems require different designs (e.g., factorial design for interaction effects). Adequate planning helps efficient observation and appropriate inferences.
    - **To control variance under study**: Score deviation is variance, and it must be controlled.
        - **Principle 1: Maximize the variance of the variable (Experimental Variance)**: Main task is to maximize the systematic variance of the dependent variable influenced by the independent variable.
        - **Principle 2: Control extraneous variance**: Minimize extraneous variables confounding results.
            - Ways to control: **eliminate** the variable, control through **randomization**, control through **statistical methods**.
        - **Principle 3: Minimize error variance**: Error variance is unpredictable (e.g., individual differences, intelligence); adversely affects findings.
            - Ways to minimize: **reduction of measurement errors** through controlled conditions, increase in **reliability of measures**.
- **Criteria for a Good Research Design**

    - Gives **specific answers** to research questions.
    - **Adequately tests the hypothesis**.
    - Presents the **appropriate question/problem**.
    - **Adequately controls extraneous independent variables**.
    - Allows **generalization of results** to other subjects.
    - Provides **internal and external validity**.
- **Types of Experimental Research Designs** (Pure experimental research often not possible in behavioral/social sciences)

    - **Single Case Experimental Design**:
        - Useful in **clinical research**, especially behavior modification.
        - Provides **detailed information of human behavior** not possible in group designs.
        - Allows **repeated measurements** and noting subtle changes.
        - **Not suitable for generalizing findings to larger populations** (small number of non-randomly selected subjects).
    - **Quasi-Experimental Design**:
        - Situations where researcher **does not have full control over random assignment** of experimental units or manipulation of treatment.

        - Limitations, but advantages: possible to select subjects randomly (e.g., students from a class), conducted in **natural/real-life settings**, providing **realism** and some **generalisability**.
        - Can provide answers to problems about past situations or those unmanageable by pure experimental design.
    - **Experimental Design**:
        - Generally conducted in the **laboratory with complete control over all variables and all subjects**.
        - Allows **random assignment** to treatment/control groups, and **manipulation of IV** to study pure effects.
        - Complete control over scheduling of IVs.
        - Allows use of **high-level advanced statistical methods**.
        - **Types of designs within Experimental Design**:
            - **Between-subjects design**: Each subject observed under **only one** of several treatment conditions.
            - **Within-subject design** (Repeated Measures Design): Each subject observed under **all** treatment conditions.
            - **Mixed design**: Combines factors from both between-subjects and within-subjects designs.
            - **Classical pretest-posttest design**: Total population randomly divided into control and experimental samples. Only experimental sample exposed to manipulated variable. Compare pre-test with post-test results for both.
            - **Solomon four-group design**: Sample randomly divided into four groups (two experimental, two control); offers more control than classical design, specifically for pretest effects.
            - **Factorial design**: Similar to classical, but uses additional samples, with each group exposed to a **different experimental manipulation**.

Unit 4: Case Study

- **Introduction**

1. Important tool in social sciences, applicable to single subjects, small/large groups, or events.
2. Involves an **in-depth, longitudinal examination of a single subject or event**.
3. Can be **descriptive or explanatory**.
4. Differs from survey: intensive study of single subject/event rather than few factors from many units.
5. Limited in scope but more exhaustive and informative than surveys.
6. Example: Teacher studying causes of failure in math for a small group of students for remedial measures.

- **Nature of Case Study**

1. Provides a **systematic and scientific way of perceiving/examining events, collecting data, analyzing information, and preparing a report**.
2. Aims for a **sharpened understanding** of why an instance happened and what to look at in future research.
3. Lends itself to both **generating and testing hypotheses**.
4. A **research strategy, an empirical inquiry that investigates a phenomenon within its real-life context**.
5. Can include quantitative evidence, relies on multiple sources, and benefits from prior theoretical propositions.
6. Single-subject research provides statistical framework for quantitative case-study data.
7. Historically used in clinical psychology for patient history and accurate diagnosis (e.g., Sigmund Freud's psychoanalysis).
8. Guidance counselors, social workers use it for diagnosing problems and recommending remedies.
9. Based on **reality** (e.g., school environment behavioral problems).
10. Uses observation, interviews, psychological tests.
11. **Subjective bias is a constant threat** to objective data gathering and analysis.
12. Typically a **single-case design**, but can be **multiple-case design** where replication is the criterion for inclusion.
13. Must provide **valid and reliable results**.

- **Criteria for Selection of Case Study**

    1. Often uses **information-oriented sampling**.
    2. Cases are based on information from **extreme or typical cases**, as average cases are often less informative.
    3. Extreme/atypical cases reveal more information by activating basic mechanisms.
    4. Random samples emphasizing representativeness seldom provide this insight.
    5. More appropriate to select a few cases for their validity.
    6. **Types of information-oriented cases**:
        - **Critical cases**.
        - **Extreme or deviant cases**.
        - **Paradigmatic cases**.
    7. Researchers decide between single-case or multiple-case studies, and whether to keep the case holistic or have embedded sub-cases.
- **Types of Case Study** (Four types)

    1. **Illustrative Case Studies**: Primarily **descriptive**, use one or two instances to show what a situation is like. Make the unfamiliar familiar and provide common language.
    2. **Exploratory (or pilot) Case Studies**: Performed **before a large-scale investigation**. Helps identify questions and select measurements for the main study. Risk of releasing initial findings prematurely as conclusions.

3. **Cumulative Case Studies**: Aggregate information from **several sites collected at different times**. Aims for greater generalization without new, repetitive studies.
4. **Critical Instance Case Studies**: Examine one or more sites to study a situation of **unique interest** (little interest in generalizability) or to **challenge a highly generalized assertion**. Useful for answering **cause-effect questions**.

- **Steps for Case Study**

1. **Determining the present status of the case or cases**: Through direct observation, physical examination, and psychological evaluation (e.g., for a 'slow learner').
2. **Identifying the most probable antecedents of the case or cases**: Helps formulate a workable hypothesis (e.g., slow learning due to unhealthy home environment, bad study habits, poor teaching).
3. **Verification of Antecedents/Hypotheses**: Check for presence/absence of supposed antecedents using a **multi-method approach** (observation, past history, interview).
4. **Diagnosis and Remedial Measures**: Diagnose the causes and suggest remedial measures.
5. **Follow-up of the case or cases**: Study the impact of remedial measures; if positive, diagnosis is considered correct.

- **Ways of Case Studies** (Different ways of using case studies)

1. **Writing analysis of case study**: Most careful analysis, can be used as term papers with readings/bibliographies.
2. **Panel of experts**: Listening to experts discuss a case is useful as an introduction to the method.
3. **Analysis of similar case studies**: Collecting similar incidents from group members' experience to draw generalizations.
4. **Cross-examination**: Group members answer pre-prepared questions, requiring careful thinking and preparation, and defending points of view.

- **Misconception about Case Study** (Flyvbjerg, 2006, identifies five)

1. Theoretical knowledge is generally more valuable than concrete, practical knowledge; thus, **case study cannot contribute to scientific development** because one cannot generalize from an individual case.
2. Case study is most useful **only for generating hypotheses**, while other methods are better for hypothesis testing and theory building.
3. The case study may be affected by a **bias tendency toward verification** (confirming researcher's preconceived notions).

Here are good notes for your exam, arranged by Block, Unit, and Topic, drawing on the provided sources.

Block 3: Research Design

Unit 1: Single Factor Design

- **1.0 Introduction**
    - **Research** is a systematic way of thinking to develop and test new theories and enhance knowledge, striving to be unbiased and objective.
    - **Research design** is a crucial step in planning a research study, explaining how answers to research questions will be found, enabling valid findings and conclusions.
- **1.2 Meaning of Research Design**
    - A **research design** is the **plan, structure, and strategy** of an investigation, conceived to answer research questions and control variance.
        - **Plan:** The overall scheme, from hypothesis writing to data analysis.
        - **Structure:** The outline or paradigm of variable operation.
        - **Strategy:** Methods for gathering and analyzing data to achieve objectives and tackle problems.
    - Essentially, it's a **blueprint** or **detailed plan** for how a research study will be completed, including how variables are measured, samples selected, data collected, and results analyzed.
- **1.3 The Function of a Research Design**
    - To **provide answers to research questions**.
    - To enable the researcher to answer questions **validly, accurately, and economically**.
    - To **control variance**.
        - This is guided by the **MAXMINCON principle**:
            - **Maximise systematic variance:** Variability in the dependent measure due to independent variable manipulation.
            - **Control extraneous variance:** Influence of other variables on the dependent variable, controlled by methods like randomisation, elimination, or matching.
            - **Minimise error variance:** Variability due to variables uncontrollable by the experimenter.
- **1.4 Basic Terminology in Research Design**
    - **Factor:** The independent variable of an experiment; there is always at least one.
    - **Levels:** A particular value of an independent variable; an independent variable has at least two levels.
        - *Example:* If studying the effect of **reward** on verbal learning, "reward" is the **factor**, and its levels are "**reward**" or "**no reward**".
    - **Treatment:** A particular set of experimental conditions.

        - *Example:* In a 2x2 factorial experiment, subjects are assigned to 4 treatments (combinations of levels).
- **1.5 Single Factor Design**
    - Used when there is **only one independent variable**.
    - Classified into two categories: **Between Group Design** and **Within Subject Design**.
    - **Between Group Design:**
        - Subjects are **randomly assigned to different treatment conditions**.
        - **i) Two Randomized Group Design:**
            - Subjects are randomly assigned to two groups, typically an **experimental group** and a **control group**.
            - *Example:* A researcher wants to see the effect of **knowledge of result** (independent variable) on the **rate of learning** (dependent variable) in school students. 100 students are randomly divided into 50 experimental (receive knowledge of result) and 50 control (no feedback).
            - Common statistical techniques: **'t' test** or **Mann-Whitney U test**.
        - **ii) More than Two Randomized Group Design (Multigroup Design):**
            - Used when there are **more than two or three experimental groups**, sometimes with a control group.
            - *Example:* Studying the effect of **schedule of reinforcement** on verbal learning, with three experimental groups receiving different reinforcement schedules and one control group receiving none.
            - Common statistical techniques: **one-way analysis of variance** and **Duncan Range test**.
        - **iii) Matched Group Design (Randomised Block Design):**
            - Subjects are **first tested on a common task** and then formed into **equivalent groups** (matched groups).
            - The **matching variable** should have a high correlation with the dependent variable.
            - *Example:* To study the effect of **praise** on intelligence test performance, subjects are initially tested on Form A of an intelligence test, paired based on scores, and then randomly divided into two groups (one praised, one not) for Form B.
            - Matching methods: **Matching by pairs** (each person in group 1 has a match in group 2) or **matching groups in terms of Mean and Standard Deviation** (when Conditions Within Subject Design:** Simplest, all subjects experience both conditions (e.g., comparing reaction time for red vs. green).

        - **Multiple Conditions Within Subject Design:** Used when more than two conditions are compared or to determine the function relating independent and dependent variables.
            - *Example:* Fergus, Craik, and Tulving (1975) examined if different processing strategies (visual, acoustic, semantic) affect memory. Subjects experienced all three strategies, and recognition rates were compared (e.g., 18% visually processed, 70% acoustically, 96% semantically).
    - **Controlling for Order and Sequence Effects:**
        - **Order effects** result from the ordinal position of a condition.
        - **Sequence effects** depend on interaction between specific conditions (e.g., a light weight feeling lighter after a heavy one).
        - Control methods:
            - **Randomisation:** When each condition is given multiple times to each subject (e.g., BCAD, ADCB for four conditions presented twice).
            - **Reverse counterbalancing:** When fewer subjects or conditions are presented few times (e.g., ABC, CBA sequence).
- **Comparison Between Group and Within Subject Design**

    - **Within Subject:** Each subject receives **multiple treatments**. Increases precision by eliminating inter-subject differences as an error source.
    - **Between Subject:** Each subject receives **only one treatment**.
    - **Preference:** Within-subject design is preferred with a **small number of subjects** available for extended experimentation and a small number of treatments.

Unit 2: Factorial Design

- **2.0 Introduction**

    - Used when there are **more than one independent variable with more than one level**.
    - Allows studying the **interaction** of two or more variables, in addition to their main effects, providing more complete information than single factor designs.
- **2.2 Meaning of Factorial Design**

    - A design where **two or more variables (factors) are employed** such that **all possible combinations of selected values of each variable are used**.
    - Enables the study of both the **independent (main) effect** and **interactive effect** of multiple independent variables.
- **2.3 Terms Related to Factorial Design**

    - **Factors:** The independent variables (manipulated or selected).

        - *Example:* Drug dosage (manipulated) and age (selected) can both be factors.
    - **Main Effect:** The **effect of one independent variable averaged across all levels of another independent variable**.
    - **Interaction:** Occurs when the **effect of one independent variable depends on the level of another independent variable**.
        - *Example:* A diet (B) combined with exercise may yield a "bonus" effect beyond the sum of their individual effects, indicating an interaction.
    - **Types of Interaction:**
        - **Antagonistic interaction:** Main effect is non-significant, but interaction is significant; variables tend to reverse each other's effect.
        - **Synergistic interaction:** Higher level of one independent variable enhances the effect of another.
        - **Ceiling effect interaction:** Higher level of one independent variable reduces the differential effect of another.
    - **Randomisation:** Process of allocating experimental units to treatment randomly, ensuring each treatment has an equal chance of being applied to each unit.
    - **Blocking:** Grouping experimental units into homogeneous clusters to improve treatment comparison.
- **2.4 Simple Two Factor Design**

    - **2x2 Factorial Design:** Two independent variables, each with two levels.
        - *Example:* Factor A (A1, A2) and Factor B (B1, B2) create four treatment combinations (A1B1, A2B1, A1B2, A2B2).
    - **2x3 Factorial Design:** Two independent variables, one with two levels and one with three.
    - Most experiments use two or three factors with two to six levels.
    - **Representing Interaction in Graphic Form:**
        - If lines representing conditions on a graph are **not parallel** (diverge, converge, or cross), it suggests an **interaction**.
        - Parallel lines suggest an additive relationship (no interaction).
        - When interaction is present, **main effects can be misleading**; it's important to examine **simple effects** (effect of one factor at each level of the interacting factor).
- **2.5 Types of Factorial Design**

    - **Within Subject Factorial Design: Each subject experiences all experimental conditions** (all combinations of factors).
        - *Example:* Godden & Baddeley (1975) memory study where deep-sea divers learned and recalled words in four conditions (beach-beach, beach-underwater, underwater-beach, underwater-underwater). All divers experienced all four.
    - **Between Subject Factorial Design: Separate groups of subjects experience each condition** (each combination of factors).

        - *Example:* A 2x2 design would require four distinct groups of subjects, one for each of the four experimental conditions.
    - **Mixed Factorial Design:** Has **at least one within-subject variable and at least one between-.
    - Results are **more comprehensive and generalisable** due to multiple variable manipulation in one experiment.
- **2.7 Limitation of Factorial Design**

    - Experimental setup and statistical analysis can become **very complicated** with more factors (especially three or more with multiple levels).
    - Difficult to select a **homogeneous group** when the number of treatment combinations is large.

Unit 3: Quasi Experimental Design

- **3.2 Meaning of Quasi Experimental Design**

    - "Quasi" randomisation to try and create comparable groups.
        - *Example:* Studying gender differences in verbal learning figures; researchers select groups of males and females, as gender cannot be randomly assigned or manipulated.
- **3.3 Difference Between Quasi Experimental Design and True Experimental Design**

    - **True Experimental:** Experimenter has **complete control** over the experiment, including **random assignment** of subjects to conditions and manipulation of variables.
    - **Quasi Experimental:** Experimenter **lacks control over random assignment**. Manipulation of variables is **not possible group equivalence.
    - **3.4.2 Non-Equivalent Control Group Design:** Compares an experimental and control group, but groups are chosen by **convenience** (not randomisation).
        - *Diagram:* G1 O1 X O2, G2 O3 O4 (O=Observation, X=Treatment). Problem: Cannot definitively attribute differences to the training program due to potential pre-existing differences.
    - **3.4.3 The Separate Pretest-Posttest Sample Design:** Pretest subjects are **not the same** as posttest subjects. Useful when measuring the same individual pre-post is not possible (e.g., different customers over time).
    - **3.4.4 The Double Pretest Design:** Includes **two pre-program measures**. Strong for internal validity as it can detect if groups are maturing at different rates prior to the program, controlling for selection maturation threats.
    - **3.4.5 The Switching Replications Design:** Two groups, three measurement phases. One group gets treatment, then serves as control while the other gets treatment. Strong internal validity, may enhance external validity.
    - **3.4.6 Mixed Factorial Design:** Contains at least one **between-subject variable** and at least one **within-subject variable**.

        - *Example:* Studying the effect of **gender** (between-subject) and **caffeine intake** (within-subject, as participants experience both caffeinated and decaffeinated conditions) on pain perception.
    - **3.4.7 Interrupted Time Series Design:** Compares the **same group over time** by analyzing data trends before and after an experimental manipulation (treatment). Multiple pre-tests and post-tests.
    - **3.4.8 Multiple Time Series Design:** Similar to interrupted time series but includes **two groups** (one receives treatment, the other does not). Adding a comparison group helps clarify effects.
    - **3.4.9 Repeated Treatment Design:** A treatment is **withdrawn and then presented a second time**. Measures are taken before and after each treatment presentation and withdrawal.
    - **3.4.10 Counter Balanced Design (Cross-over design):** Experimental control achieved by **randomly applying experimental treatment** such that each treatment appears once and only once in each column and row. Controls for maturation, selection, and experimental mortality.
- **3.5 Advantages and Disadvantages of Quasi Experimental Design**
    - **Advantages:**
        - Very useful in **social science** when randomisation is difficult or unethical (e.g., studying maternal alcohol use).
        - Easier to implement in **natural settings** and allows direct assessment of treatment effects.
        - Can **reduce time and resources** as extensive pre-screening and randomisation are not needed.
        - Minimises threats to **external validity** as natural environments are less artificial than labs.
        - Findings can be generalised to other subjects and settings.
        - Useful for **longitudinal research** and measuring social variables.
    - **Disadvantages:**
        - **Low internal validity** due to lack of randomisation, making it difficult to rule out confounding variables and establish causal relationships.
        - Statistical tests can be meaningless without proper randomisation.
        - Possible **selection bias**.

Unit 4: Other Designs (Correlational Design and Comparative Design)

- **4.0 Introduction**
    - Used when researchers are interested in variables that **cannot be manipulated** (e.g., general intelligence, gender, skull size, existing diet).
- **4.2 Definition of Correlational Research Design**
    - Assumes reality is a network of **interacting and mutually causal relationships**.

- Researcher collects **two or more sets of data from the same group of subjects** to determine if a **relationship exists** between the variables.
- Studies relationships **among variables, none of which may be the actual cause of the other**.
- A **quantitative design**.

- **4.3 Types of Correlational Research Design**

  - **1) Measuring Degree of Association between Variables:**
    - **a) Association between two variables:**
      - Calculate a **correlation coefficient** (between -1.00 and +1.00).
      - **Positive correlation:** Higher on one variable, higher on the second (**direct relationship**).
      - **Negative correlation:** Higher on one variable, lower on the second (**inverse relationship**).
        - *Example:* If academic stress and academic achievement have a negative correlation, high stress means low achievement.
    - **b) Association between more than two variables:** Measures associations among multiple variables using **multiple correlation**.
  - **2) Predicting Scores on One Variable:**
    - If a correlation exists, one score can be predicted from another using **predictor variables** and **criterion variables**.
      - **Predictor variable:** Used to forecast an outcome (e.g., stress score).
      - **Criterion variable:** The variable being predicted (e.g., future health status).

- **4.4 Evaluation of Correlational Design**

  - **Advantages:**
    - Easy to use with **available data**.
    - Can **formulate new hypotheses** for more rigorous research designs.
    - Avoids **pretest sensitisation**.
    - Uses **realistic measurements** and avoids non-representative contexts.
    - Permits **large, carefully chosen samples**.
  - **Disadvantages**
  - Careful choice of variables based on **sound theory** that accounts for alternative explanations.
  - Use of statistical techniques (e.g., hierarchical multiple regression, path analysis, structural equation modeling) that allow **partial control of third variables** when measured.
  - Emphasis on **controls put in place to control extraneous variables**.

- **4.6 Causal Comparative Research Design**

- Used when the researcher is **unable, unethical, or too difficult to control the independent variable**..7.1 Similarity:**
    - Both are **non-experimental methods** because they lack manipulation of an independent variable and random assignment.
    - Variables are **observed in natural settings**.
    - Both use techniques like **matching or quota sampling** to control confounding variables.
    - Both often use **attribute variables** (characteristics that cannot be manipulated, like intelligence or gender) as independent variables.
- **4.7.2 Differences:**
    - **Causal Comparative:** Compares **two or more groups** to explain existing differences between them. Involves at least one **categorical variable**.
    - **Correlational:** Looks for **relationships within a single group**. Includes **only quantitative variables**.

- **4.8 Comparison Between Causal Comparative and Experimental Design**

    - **Experimental:** Researcher randomly assigns subjects to treatment groups and manipulates the independent variable.
    - **Causal Comparative:** Individuals are **not randomly assigned** to groups; they are already in those groups, and the independent variable cannot be manipulated by the researcher.
- **4.9 Data Analysis for Causal Comparative Research Design**

    - **t-test:** For two groups (independent t-test for independent groups, dependent t-test for matched/pre-post designs).
    - **One-way analysis of variance (ANOVA):** For three or more groups.
    - **Chi-square test:** For frequency counts of the dependent variable (e.g., comparing political affiliations).
- **4.10 Evaluation of Causal Comparative Research Design**

    - **Advantages:**
        - Useful for **behavioral science** when organismic variables (age, gender) cannot be manipulated.
        - Applicable when manipulating the independent variable is **unethical** or harmful.
        - Less costly than experimental studies.
        - Helps **identify variables worthy of experimental investigation**.
    - **Limitation:**
        - **Less control** than experimental studies because the independent variable has already occurred.
        - **Causation is difficult to establish:** The alleged cause might be the effect, or a third variable influences both (e.g., self-esteem and reading achievement could influence each other or both be influenced by intelligence).

        - Cannot randomly assign participants, so groups may differ on other variables (confounds).

Block 4: Qualitative Research in Psychology

Unit 1: Introduction Including Ethnography

- **1.2 Meaning of Qualitative Research**
    - A type of scientific research that aims to **bridge information gaps**, systematically collects evidence, and finds answers to problems or questions.
    - Widely used for **in-depth understanding of human behavior**, opinions, values, and social aspects of a community, culture, or population, particularly regarding reasons behind decision-making.
- **1.3 Types of Qualitative Research**
    - **i) Case Study:** In-depth study of an individual, group, event, institution, or society, often using multiple data collection methods (interviews, observation, documents) to provide a rich and holistic description.
    - **ii) Ethnography:** Focuses on a **particular community**, involves **close-field observation** and studies socio-cultural phenomena. Can be used for comparative analysis (Ethnology) or historical analysis (Ethnohistory).
    - **iii) Historical Method:** Helps understand and analyze **causal relationships** by collecting and evaluating data on past events to explain present and anticipate future events.
    - **iv) Grounded Theory:** Involves **active researcher participation** and observation to **generate or develop theories** based on collected data.
    - **v) Phenomenology:** Explains behavioral phenomena through the **conscious experience of events**, without relying on external theories or assumptions.
        7. *Example:* Patients describing caring nurses based on their "existential presence" (positive response to requests) rather than just physical presence.
- **1.4 Qualitative and Quantitative Research: A Comparison**
    - **Basic Difference:**
        7. **Qualitative:** Based on **observation and experiences**.
        8. **Quantitative:** Based on **numerical or graphical representation of data**.
    - **Other Key Differences:** | Feature | Qualitative Research | Quantitative Research | | :---------- | :---------- | :---------- | | **Objectives** | Explores phenomena, describes variation, explains relationships, describes behavior/experiences/norms. | Confirms hypotheses, quantifies variation, predicts causal relationships. | | **Questions** | **Open-ended**. | **Close-ended**. | | **Data** |

Notes, recordings, video tapes. | **Numbers and graphs**. | | **Design** | Flexible; questions adapted based on participant response. | Predetermined and stable; structured questions for all participants. |

- **1.5 Relevance of Qualitative Research in Psychology**
    - Helps in **textual description of people's experiences**.
    - Identifies and explains **social norms, religion, gender roles, socio-economic status**.
    - Helps understand **unquantifiable behavioral phenomena**.
    - Collects data under **more natural situations**.
    - Determines factors meaningful and important to respondents.
- **1.6 Ethnography: The Meaning**
    - Also known as **'ethnomethodology'** or **'methodology of people'**.
    - Studies culture through **close observation and active participation**, focusing on the socio-cultural phenomena of a community.
    - Uses a **"chaining process"** where participants identify more respondents, allowing for deeper responses and comprehensive data collection.
    - Ethnographers often **immerse themselves in the community for months** to collect observational transcripts and interview recordings.
- **1.7 Some Common Terms Used by Ethnographers**
    - **Symbols:** Cultural traditions or material artifacts (art, clothing, rituals).
    - **Cultural patterning:** Study of cultural patterns formed by relationships between symbols.
    - **Tacit knowledge:** Firm, deeply embedded cultural beliefs rarely discussed explicitly, inferred by the ethnographer (e.g., prejudices).
    - **Situational reduction:** Belief that social structures result from interactions of several social factors.
- **1.8 Assumptions in Ethnography**
    - Research objective depends on **community cultural understandings** (risk of overestimating culture, underestimating causal reasons).
    - Difficulty in identifying the target community and potential for researcher bias towards their own culture.
    - Assumes researcher is an **expert in the culture's norms, mores, and language**.
- **1.9 Types of Ethnographic Research**
    - **i) Macro ethnography:** Study of **broadly defined cultural groupings** (e.g., "the Indians").
    - **ii) Micro ethnography:** Study of **more specific cultural groupings** (e.g., "local government").

    - **iii) Emic perspectives:** Viewpoints of **'ingroup' members** (from within the culture).
    - **iv) Etic perspective:** Viewpoints of **'outgroup' members** (from outside the culture).
- **1.10 Purpose of Ethnographic Research**
    - Facilitates **cross-cultural analysis**.
    - Helps analyze **past events or culture history**.
    - Studies behavior, experiences, and attitudes in a **natural environment**.
    - **Close observation increases validity** of reports and theories.
- **1.11 Steps of Ethnographic Method**
    7. **Selection:** Choose a culture.
    8. **Review of Literature:** Get a historical sketch.
    9. **Identification of variables:** Identify variables of interest.
    10. **Entry:** Gain acceptance into the culture.
    11. **Cultural Immersion:** Live within the culture for extended periods, gaining informants.
    12. **Data Collection:** Gather observational transcripts and interview recordings.
    13. **Development of theory:** Formulate theory based on interpretations, seeking validation from community members.
- **1.12 Ethical Guidelines in Qualitative Research**
    - Respect respondents and their responses.
    - Show respect and belongingness to the community.
    - Inform respondents about the research's purpose.
    - Maintain confidentiality.
    - Be aware of potential risks and benefits.

Unit 2: Grounded Theory

- **2.2 Goals and Perspectives of Grounded Theory**
    - To **generate new theories or modify existing ones**.
    - Ensures a "good theory" output due to systematic steps and multiple data sources.
    - Emphasizes the **process of theory evaluation** and enhancing **theoretical sensitivity**.
    - Formulates **hypotheses based on conceptual ideas** and explains people's actions.
    - Results are **probability statements about concept relationships**, not just facts.
- **2.3 Methods of Grounded Theory**

    - Does not believe in extensive taping/transcribing interviews (considered time-wasting).
    - Relies on **field-noting interviews** and quickly generating concepts.
    - **"Everything is data"**: Any information that helps generate concepts for the emerging theory (field notes, informal interviews, lectures, media, self-interviews).
- **2.4 Steps of Grounded Theory**
    - **2.4.1 Memoing:** Collecting data in **short notes (memos)**.
        - **a) Theoretical notes:** Relate textual data to existing literature.
        - **b) Field note:** Observations and causal notes from active participation in the study setting.
        - **c) Code notes:** Discuss codes used for naming, labeling, or categorizing events.
    - **2.4.2 Sorting:** Organizing collected information (memos) in proper order to link ideas.
    - **2.4.3 Writing:** Arranging, relating, and putting information into words, interpreting data based on researcher's perspectives and linking to literature.
- **2.5 Types of Coding in Grounded Theory**
    - **2.5.1 Selective Coding:** Selecting **one central category** and relating all other categories to it.
    - **2.5.2 Open Coding:** Identifying, labeling, and analyzing phenomena in the text, categorizing them into general dimensions.
    - **2.5.3 Axial Coding:** Relating categories to each other using deductive and inductive thinking, analyzing causal relations between codes ("cause" and "context").
- **2.6 Relevance of Grounded Theory**
    - Increases **reliability and validity** of theory due to data from multiple sources.
    - Allows exploring facts and analyzing **causal reasons** behind them.
    - Is an **inductive** research approach, based on observations and data.
    - Categorizes data, forming a basis for organizing and reporting results.
- **2.7 Implications of Grounded Theory**
    - Useful in **policy and program evaluation research**.
    - Can analyze consumer demands, preferences, product positioning, and advertising opportunities.
    - Applicable in education, management, women's studies, information studies, politics, and communities.
    - Helps understand and describe **human psychology and experience**.
- **2.8 Criticism of Grounded Theory**
    - Potential for **researcher preconceptions** in data collection and analysis.

- **Content validity can be questionable** despite corpus data collection.

Unit 3: Discourse Analysis

- **3.0 Introduction**
    - "Discourse" refers to **"talk"** – the ways people describe or explain their experiences.
    - **Discourse analysis** (originating in the 1960s) explains talk or texts (e.g., historical incidents, social activities).
- **3.2 Definition of Discourse/Content Analysis**
    - Study of how versions of the world, society, events, and psyche are produced through the **use of language and discourse**.
    - **Content analysis** is a research technique for the **objective, systematic, and quantitative description of manifest content of communications**.
    - Identifies linguistic dependencies between sentences or utterances.
    - Can be seen as a **creative way of approaching a problem** or providing tangible answers based on scientific research.
    - Essentially, it's a **deconstructive reading and interpretation** of a problem or text.
- **3.3 Assumptions of Discourse Analysis**
    - Challenges the assumption of objectivity in human behavior studies, as **researcher expectations can influence interpretation**.
    - Assumes **reality is socially constructed**: language shapes the categories and constructs we use, and since language is social, our sense of reality is socially and culturally constructed.
    - Assumes **people are the result of social interaction**: many "inner essences" (personality, anxiety) are products of social interaction.
- **3.4 Approaches or Theories of Discourse Analysis**
    - **Modernism:** Views discourse as related to talking, emphasizes language transformations for new inventions/understandings, leading to discourses of rights, equality, freedom.
    - **Structuralism:** Argues human actions and social formations are related to language/discourse as systems of related elements, where individual elements gain significance in the context of the whole structure.
    - **Postmodernism (Foucault):** Investigates the **variety of individual experiences**, emphasizing differences. Views discourse as "systems of thoughts" constructing subjects and worlds, having a significant role in **power relations**. Argues **power and knowledge are inter-related**.
    - **Feminism:** Explains discourse as social practices, investigating relationships among power, ideology, language, and discourse, emphasizing **"performing gender"**.

- **3.5 Steps in Discourse Analysis**
    - Evaluates patterns of speech, viewing speech as a performance (an action rather than a description).
    - **i) Target orientation:** Define the focus of study and how data will be analyzed.
    - **ii) Evaluation:** Examine the value of collected data, especially from multiple sources.
    - **iii) Interpretation of the data:** Understand and interpret the data for clarity.
    - **iv) Analysis of the findings:** Mechanically analyze, interpret, and summarize data (software can assist with mechanical tasks but not interpretation).
- **3.6 Relevance/Implications/Significance of the Discourse Analysis**
    - Enables understanding of connotations behind **historical events and current social practices**.
    - Provides a comprehensive view of "problems" and helps understand **hidden motivations**.
    - Aids in **"deconstructing" concepts, belief-systems, or social values**.
    - Can be applied to **any text or situation** and requires **no specific guidelines**.
- **3.7 Issues of Reliability and Validity in Discourse Analysis**
    - Lacks a proper format or guideline, making data processing controversial.
    - Interpretation can involve **researcher subjectivity or bias**.
    - No "hard data"; reliability and validity depend on the **force and logic of arguments**.
    - Arguments are subject to deconstructive reading and counter-interpretations.
- **3.8 Concept of Critical Discourse Analysis (CDA)**
    - A type of discourse-based research that studies how **social power abuse, dominance, and inequality are enacted, reproduced, and resisted by text** in social and political contexts.
    - Researchers explicitly aim to realize, experience, and overcome social inequality and injustice.
    - Focuses on **powerful groups** (leaders, institutions) that control public discourse (e.g., scholarly, educational, media, legal, political discourse).
- **3.9 Implications of Critical Discourse Analysis**
    - Allows researcher to be an **"active agent"** in exposing inequality and injustice.
    - Emphasizes the **structure and social context of media texts** to "denaturalise" ideological messages.
    - Language can be used as an **interdisciplinary tool**.
    - Helps in **cross-cultural study of social activities** via media texts.
- **3.10 Concept of Content Analysis**

    - A form of discourse analysis that **summarizes any form of content after deep study**.
    - Aims to objectively evaluate and understand situations.
    - Results are often presented in **numbers and percentages**.
    - Serves two purposes: removes subjectivity from summaries and facilitates trend detection.
- **3.11 Examples of Content Analysis**
    - **Media content:** Print media, broadcasts, recordings.
    - **Audience content:** Questionnaires, interviews, group discussions.
- **3.12 Implications of Content Analysis**
    - Enables links between causes (e.g., program content) and effects (e.g., audience size).
    - Evaluates and improves media programming.
    - Increases awareness and summarizes documentaries.

Unit 4: Reporting and Evaluating in Qualitative Research

- **4.0 Introduction**
    - Research is incomplete without **detailed reports of results/achievements**.
    - Involves systematically **analyzing, evaluating, and reporting data** to provide insights and serve as a source for future research.
- **4.2 Concept and Meaning of Evaluating or Analysing of Data in Qualitative Research**
    - An attempt to understand the extent to which collected information answers the pre-planned aims and objectives of the research.
    - Involves finding solutions or conclusions by systematically analyzing and organizing gathered information.
    - Provides insight into problems or goals.
- **4.3 Steps of Evaluating or Analysing of Data in Qualitative Research**
    - **i) Reading the overall collected data:** Go through all details from various sources (interviews, tapes, observations).
    - **ii) Categorising the collected data:** Sort relevant information and group similar data into themes (e.g., suggestions, experiences).
    - **iii) Naming or labeling the categories:** Assign labels to the sorted and categorized themes.
    - **iv) Identification of the causal relationships:** Discover patterns, associations, and causal relationships among the labeled themes based on responses and researcher experience.
    - **v) Recording or filing the data:** Maintain records of analyzed relationships for future reference.

- **4.4 Concept and Meaning of Interpreting of Data in Qualitative Research**
    - **Summarizing the findings** of data analysis to provide useful information related to research goals.
    - Involves presenting findings as a viewpoint, comparing them with expectations, or conducting a **SWOT analysis**.
    - Interpretations should be justified and supported by relevant literature.
- **4.5 Strategies of Data Interpretation in Qualitative Research**
    - **i) Making a final list:** Summarize labeled categories and codes into a list of findings.
    - **ii) Elaborate narratives:** Provide detailed explanations of findings from interviews, recordings, and discussions.
    - **iii) Use of matrices:** Use charts (like cross-tables) to compare different groups or datasets on important variables using keywords.
        - *Example:* A matrix showing age group and gender for a cosmetic product.
    - **Flow chart:** A diagrammatic representation of information flow.
- **4.6 Concept and Meaning of Reporting of the Data in Qualitative Research**
    - The ultimate discussion of interpretations and findings, supported by relevant literature and reasons. It's the final challenge for the researcher.
- **4.7 Steps of Preparing a Report of the Data in Qualitative Research**
    - **i) Preparation of the content of report:** Formulate findings according to the needs of the intended audience (e.g., funders, clients).
    - **ii) Review and discussion of the report:** Discuss interpretations and results with the people on whom the research was conducted.
    - **iii) Preparation of the executive summary:** A concise summing up of conclusions and recommendations, possibly including details of questionnaires used.
    - **iv) Scope of future research:** A list of areas for further study, acting as a research plan.
- **4.8 Contents of Research Report**
    - **1) Title Page:** Topic, organization, date.
    - **2) Table of Contents:** List of sections with page numbers.
    - **3) Executive Summary:** One-page overview of findings and recommendations.
    - **4) Purpose of the Report:** Aims, objectives, research type.
    - **5) Contextual background:** Historical background, problem, goals, literature review.
    - **6) Methodology:** Sample, scales used, data type (interviews, observations).
    - **7) Results and findings:** Analysis of collected data.
    - **8) Interpretation and Conclusion:** Discussion of findings and their significance.
    - **9) Limitations of the study:** Restrictions on generalizability.

    - **10) Recommendations and implications:** Suggestions for action.
    - **11) References:** Acknowledging sources.
    - **12) Appendices:** Supporting documents (questionnaires, forms, case studies).
- **4.9 Do’s and Don’ts in Evaluating, Interpreting and Reporting Data in Qualitative Research**

    - **Do's:**
        - Be **clear and specific** about sample populations.
        - **Code the data** for convenience.
        - Use **diagrams, flow charts, or matrices** frequently.
        - Draw conclusions based on present and related studies.
        - Develop policies to confirm qualitative data validity.
    - **Don'ts:**
        - **Don't depend fully on the selected research design** (no design is perfect).
        - **Don't interview only about successes**; failures can provide significant information.
        - **Don't completely depend on questionnaires**; observations and interviews provide richer data.